Chapter 10

Incorporation of robustness and adaptiveness into reservoir operations under climate change

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10.1 Introduction

Recently observed data have proven that global warming is at least partially due to anthropogenic causes. Despite global efforts to mitigate the impact of climate change on water resources management, this problem shows no immediate signs of abatement. Therefore, appropriate strategies should be devised to mitigate the impacts of climate change.

Adapting to changing climate has been a new challenge that is beyond the realm of conventional decision-making theories. The inability of these methods comes from the fact that climate change is a non-stationary process with large or deep uncertainties. Note that a time series is non-stationary when its key statistics (such as mean and variance) are a function of time, whereas deep uncertainty implies that upcoming future events and their likelihoods of occurrence are unknown (Kim & Chung, 2017). For example, it has been observed that the annual mean temperature of the Earth has been rising (i.e., indicating non-stationarity) while its future long-term projections, which often become the key inputs to decision-making processes, are all heterogeneous across space and time (i.e., indicating deep uncertainty). For the last two decades, many studies have addressed the importance of considering the uncertainty and non-stationarity of climate change for adaptation strategies (Buurman et al., 2009; Dessai et al., 2009; Zhang & Bobovic, 2011; Steinschneider & Brown, 2012; Park et al., 2014; Herman et al., 2015; Buurman & Babovic, 2016).

Most of these studies concluded that decision-making under climate change should be "robust" and "adaptive." A "robust" alternative should perform satisfactorily over a wide range of possible scenarios, whereas an "adaptive" alternative should be iteratively updated as new information is available (Kim & Chung, 2017). The word "flexible," which leaves several options open (Walker et al., 2013), has been widely used as a substitute for both words, "robust" and "adaptive" (Kim & Chung, 2017).

Water resources planning and management, which often require decision-making theories, have been severely affected by climate change; hence, it should also adopt robust and adaptive concepts. In this chapter, two decision-making theories that have been recently applied to the water sectors for climate change adaptation are introduced: robust optimization for the robust perspective and real option analysis (ROA) for the adaptive perspective. As case studies, this chapter then deals with a reservoir operation problem for the robust optimization and with a drought mitigation project planning for the real option analysis.

10.2 Methodology

10.2.1 Robust approach

This section mainly focuses on the robust optimization theory followed by a brief introduction of the Robust Decision Making (RDM), which was first proposed as a systematic framework for the robust perspective to handle the deep uncertainty of climate change.

10.2.1.1 Robust decision making

The key concept of RDM is to evaluate the optimal decision under multiple sets of future conditions, or "scenarios" (Matrosov et al., 2013), and to identify under which circumstances the system undergoes failure. Then, additional candidate strategies that would perform better under the conditions that cause the system to fail with the initial decision are selected and evaluated with various measures. The final step of RDM is to check, through discussions with appropriate stakeholders, whether the candidate strategies satisfy the minimum requirements; this step is conducted until the stakeholders agree with the decision-making on the new candidate strategy. The stakeholders may be provided with trade-off summaries of original and candidate strategies in the decision-making processes (Lempert & Groves, 2010).

A general RDM procedure relies on one or more of three key components: (1) multiple scenarios of plausible future climates (Lempert, 2003), (2) a robust criterion, and (3) one of the following approaches: (i) Multi-Criteria Analysis (MCA) or Multi-Criteria Decision Analysis (MCDA) under complete uncertainty (Ranger et al., 2010; Chung & Kim, 2014; Kim & Chung, 2014, 2015), (ii) assessed risk of policy (Lempert et al., 2004; Dessai, 2005; Dessai & Hulme, 2007), (iii) vulnerability and robust response (Lempert et al., 2013; Bloom, 2014) and decision scaling.

RDM approaches in water resources may vary, mainly because "robustness" can be interpreted differently by stakeholders. Lempert & Groves (2010) suggested a method to incorporate adaptive strategies into the RDM framework to aid decision-makers in designing and evaluating robust strategies. Herman et al. (2015) suggested a taxonomy of robustness framework, which includes the following: (i) identifying alternatives, (ii) sampling states of the world, (iii) quantification of robustness measures, and (iv) identification of key uncertainties (or robustness controls) using sensitivity analysis. RDM is often used collaboratively with Info-Gap Decision Theory (IGDT) (Ben-Haim, 2001) because they share a similar philosophy, which is a non-probabilistic approach to maximize robustness, with the system satisfying the minimum performance requirements. Matrosov et al. (2013) compared RDM with IGDT by applying both methodologies to London's water supply expansion in the Thames Basin.

10.2.1.2 Robust optimization

Robust optimization has been actively studied since Mulvey et al. (1995) proposed a concept that integrates goal-programming formulations with a scenario-based description of problem data. For illustration, let us consider the following simple optimization problem:

$$\text{Minimize } z = \mathbf{c}^{\mathbf{T}}\mathbf{y} + f(\mathbf{x}) \tag{10.1}$$

subject to

$$\mathbf{Ay = b} \tag{10.2}$$

$$g(\mathbf{x}) + \mathbf{By} = \mathbf{d}, \quad \mathbf{x}, \mathbf{y} \geq 0 \tag{10.3}$$

where z = the value of the objective function; $\mathbf{x}$ = a vector of control variables; $\mathbf{y}$ = a vector of decision variables; $\mathbf{c}$ = a vector of cost coefficients; $f(\mathbf{x})$ = the operation cost of the system; $\mathbf{A}$ = a matrix for the structural constraint of the model; $\mathbf{b}$ = the corresponding right-hand-side (RHS) vector of the structural constraint; $g(\mathbf{x})$ = a function for the response of the system to the value of the control variables; $\mathbf{B}$ = a matrix for the effects of the structural decisions on the system performance; and $\mathbf{d}$ = the corresponding RHS vector (Watkins & McKinney, 1997).

The robust optimization allows some violations on the second constraint of Equation (10.3) to be more "flexible" over a scenario set $\Omega = \{s_1, s_2, \ldots, s_s\}$. The formulation of the robust optimization model then becomes (Watkins & McKinney, 1997)

$$\text{Minimize } \sigma_v(\xi_1, \ldots, \xi_s) + \omega p(v_1, \ldots, v_s) \tag{10.4}$$

subject to

$$\mathbf{Ay = b} \tag{10.5}$$

$$\mathbf{g}(\mathbf{x}_s) + \mathbf{B}_s\mathbf{y} + v_s = \mathbf{d}_s \tag{10.6}$$

$$\mathbf{x}_s, \mathbf{y} \geq 0, \quad s \in \Omega \tag{10.7}$$

where the subscript s represents the index number of each possible scenario. Equation (10.4) includes both $\sigma_v(\xi_1, \ldots, \xi_s)$, an aggregate objective function containing information about the performance of the solution under all possible scenarios, and $p(v_1, \ldots, v_s)$, a feasibility penalty function that penalizes constraint violations under all possible scenarios. The additional term v_s in Equation (10.6) represents the measure of the extent of violation of the control constraint in Equation (10.3) in the robust optimization model. Watkins & McKinney (1997) referred to the second term of Equation (10.4) as feasibility robustness because one can incorporate the robust concept into another term such as the objective function. In the field of water resources planning and management, robust optimization has been actively studied for the last few decades (e.g., Ray et al., 2014; Pan et al., 2015).

10.2.2 Adaptive approach

Adaptability enables a system to change proactively to changes in the environment (De Neufville & Scholtes, 2011). In traditional decision-making situations, adjusting alternatives with future conditions has been seldom applied, because it requires considerable effort, money, and time. From the "adaptive" perspective, decision-makers can refine their decision iteratively by continuously learning from previous outcomes and new information (Kim & Chung, 2017). With this process, adaptive decision-making methods can adapt to changing climate and absorb a wide range of scenarios. Among decision-making methodologies, ROA is a well-known decision-making methodology tackling climate change.

10.2.2.1 Real option analysis

ROA spreads risks over "time" using "options." Although ROA was initially coined by Myers (1977) for the financial market, it has received attention from decision-makers in engineering since De Neufville (2003) revisited the concept and its benefits with an emphasis on flexibility under deep uncertainty in the planning of various engineering projects (Kim & Chung, 2017). Furthermore, ROA has recently begun to receive attention in the field of water resources management and planning (e.g., Kjaerland, 2007; Borison et al., 2008; Gersonius et al., 2012; Deng et al., 2013; Jeuland & Whittington, 2014; Ryu et al., 2017; Erfani et al., 2018; Ihm et al., 2019). These previous studies demonstrated how ROA becomes attractive to decision-makers tackling uncertainty issues, such as climate and socio-economic factors, in an adaptive manner.

ROA allows delaying commitment to large, costly, and irreversible decisions while exercising different interventions until more information is available (Erfani et al., 2018). Thus, "adaptation" is enabled because ROA provides an optimal sequence of future decisions that respond to changes in uncertainty over a planning horizon. There are three main types of real options associated with investment projects to operate each alternative differently; for instance, one can abandon, delay, or carry out the alternative. Such different operations of an alternative are called "real options," which are not an obligation but a right (Kim & Chung, 2017).

ROA was proposed as an alternative to a traditional benefit–cost analysis (BCA) called the discounted cash flow (DCF) approach. DCF does not allow altering a selected alternative over time, but this can be achieved through ROA over the planning horizon. As mentioned above, ROA considers uncertainty as a metric of probability. Table 10.1 summarizes the differences between DCF and ROA.

Table 10.1 Differences between DCF and ROA

	DCF	*ROA*
Uncertainties	X	O (please define the symbols X, O)
Options	X	O
Evaluation metric	Net present value (NPV)	Expanded net present value (= NPV + OP)

O, taken into consideration; X, not taken into consideration.

Within the BCA, the net present value (NPV) is defined as the difference between the present monetary value of benefits and costs. Conversely, the expanded net present value (ENPV) is defined as the value reflecting the "flexibility" of the options (Trigeorgis, 1993), which equals the sum of NPV and the "option premium (OP)." Therefore, alternatives judged to be economically infeasible in DCF can be evaluated as feasible in ROA, reflecting flexibility (Ihm et al., 2019) because of the OP. The ENPV and OP are calculated by Equations (10.8) and (10.9), respectively, as follows:

$$\mathrm{ENPV}_t = \max\left[\mathrm{NPV}_t, \mathrm{OV}_t^1, \mathrm{OV}_t^2, \ldots, \mathrm{OV}_t^m\right]\frac{1}{2} \tag{10.8}$$

$$\mathrm{OP}_t = \mathrm{ENPV}_t - \mathrm{NPV}_t \quad \begin{cases} \text{if} & \mathrm{ENPV}_t > \mathrm{NPV}_t \\ \text{otherwise} & 0 \end{cases} \tag{10.9}$$

where OV_t^m is the sum of the net benefits of option m at time step t, and OP_t is the OP at time step t.

10.2.2.2 Decision tree approach

ROA is implemented through different procedures, including decision trees, lattices, and Monte Carlo analysis (Chow & Regan, 2011; De Neufville & Scholtes, 2011; Lander & Pinches, 1998; Trigeorgis, 1996). Among them, the decision tree approach is the most widely-used procedure for ROA. This is because it not only considers a full range of option types but is also conceptually easy to explain and understand. To implement the decision tree model, a tree that contains real options and involves uncertainties should be assigned. Figure 10.1 illustrates an example of a simple decision tree. Though this figure is a simple example of the binomial tree, it can be extended so that the full range of option types (yellow squares in Figure 10.1), and the corresponding uncertainties (blue triangles in Figure 10.1) can be addressed.

Figure 10.2 presents the binomial tree configuration and the value propagation, which requires three parameters, u (upward), d (downward), and p (risk-neutral probability), as described in Equations (10.10–10.12).

$$u = e^{\sigma_v \sqrt{\delta t}} \tag{10.10}$$

$$d = e^{-\sigma \sqrt{\delta t}} = \frac{1}{u} \tag{10.11}$$

$$p = \frac{e^{r \cdot \delta t} - d}{u - d} \tag{10.12}$$

where σ_v is the volatility, which is the standard deviation of the natural logarithm of the underlying free cash flow return, δt is the time associated with each time step, and r is the risk-neutral interest rate.

In Figure 10.2a, each node represents a possible pathway at each discrete time step. Figure 10.2b shows the calculation of the option values using the three estimated

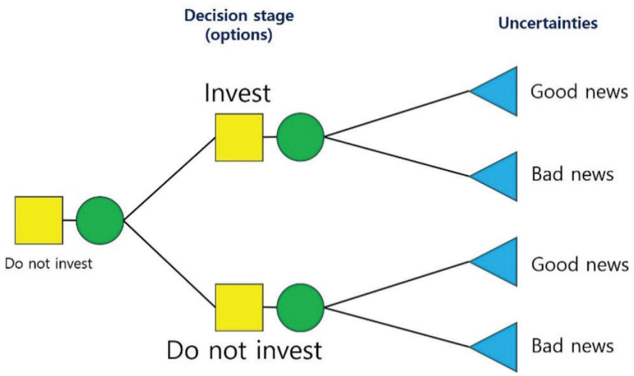

Figure 10.1 Simple example of a decision tree for ROA (Ihm et al., 2019).

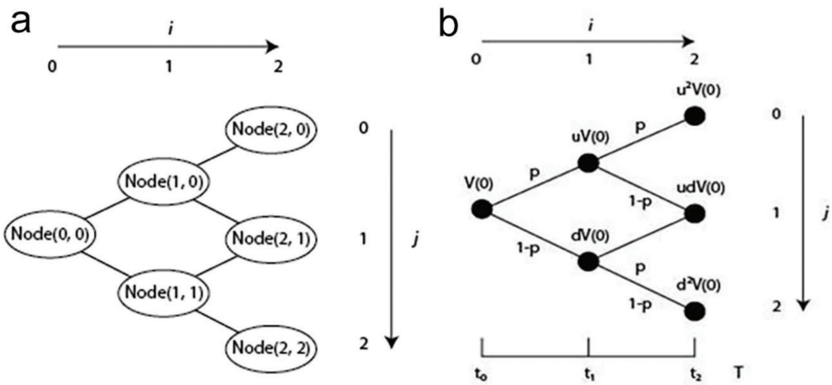

Figure 10.2 Model configuration and value propagation of the binomial tree approach (Ryu et al., 2017). (a) Node and (b) value ($i = 0, 1, 2, \ldots, N$ and $j = 0, 1, 2, \ldots, i$).

parameters: u, d, and p. The option values, $V(i,j)$, can then be calculated. As $V(i+1,j) = u \cdot V(i, j)$ and $V(i+1, j+1) = d \cdot V(i, j)$, all the values of $V(i, j)$ can be serially calculated from the initial value $V(0, 0)$ $(=V(0))$ using the forward-moving process (Ryu et al., 2017). Figure 10.3 demonstrates an example of an "option matrix," which is a form of the binomial tree model, evaluated by the three options on each node.

10.3 Case study

10.3.1 Study basin: Boryeong Dam

Boryeong Dam is located in the city of Boryeong in the South Chungcheong province, South Korea. The dam supplies fresh water to eight neighboring districts (Boryeong, Cheongyang, Dangjin, Hongseong, Seocheon, Seosan, Taean, and Yesan) through a regional water supply system (as shown in Figure 10.4). Since 2014, this region has suffered from a persistent drought for multiple years. For example, the total annual

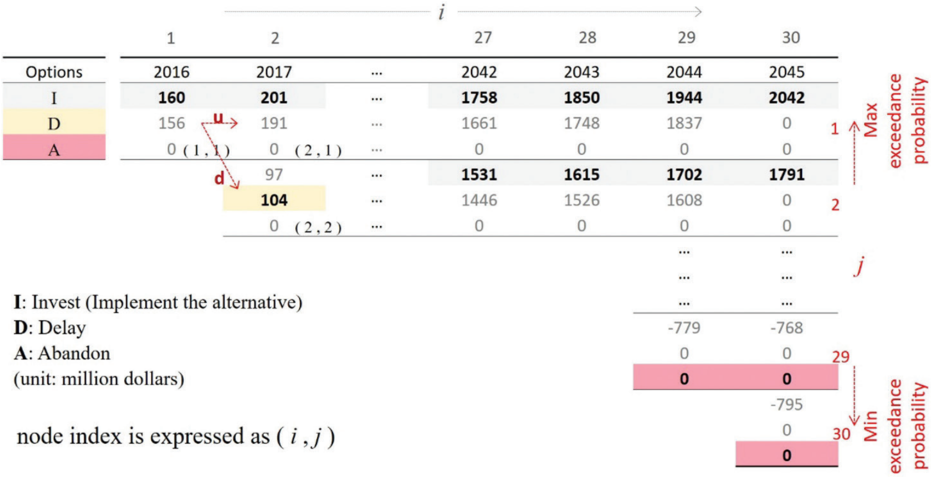

Figure 10.3 Example of the "option matrix" evaluated by the real options on each node for an alternative (Ryu et al., 2017).

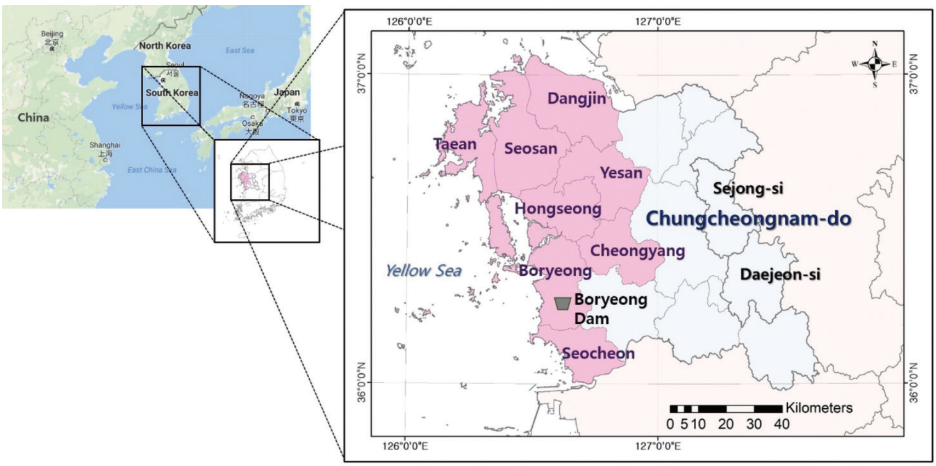

Figure 10.4 Boryeong Dam and the eight administrative districts to which Boryeong Dam supplies water (Ihm et al., 2019).

rainfall was approximately 800 mm in 2015, which is only 60% of the annual mean rainfall. Owing to the severe and persistent drought in this region, the reservoir stage remained below the severe warning level for 135 days. Consequently, the government imposed a water supply restriction from October 2015 to February 2016, which led to a massive shortage of drinking water for approximately 480,000 people.

The reservoir capacity of Boryeong Dam is not sufficiently large to store sufficient water to cope with an extreme drought, such as the multi-year drought event mentioned above. Therefore, the reservoir operation rule should be optimized to reduce the water shortage risk, which is discussed through the robust perspective in Section 10.3.2. Furthermore, the Korean government carried out a project for the construction of a water conduit (called the Conduit Project hereafter) that supplies additional water to Boryeong Dam from upstream (Ihm et al., 2019). The construction cost and operational expenses are estimated at 58 million USD and 1 million USD per year, respectively (KDI, 2016). Although the Conduit Project could reduce the risk of depletion of reservoir storage, it has been controversial in terms of economic feasibility due to its short period of construction. Therefore, Section 10.3.3 re-evaluates the economic feasibility of this project through an adaptive perspective.

10.3.2 Dam operations with robust optimization

The first case study linked the robust concept to the monthly operations of a single reservoir of the study basin (i.e., the Boryeong Dam). A conventional stochastic dynamic programming optimization model (CSDP hereafter) was first developed for the dam and then extended to the proposed robust-SDP model (RSDP hereafter) to test a mathematical form of the robustness of the SDP objective function. The monthly releases according to the CSDP and the RSDP models were calculated using "the historical inflow series," which is based on the historical mean (μ_0) and standard deviation (σ_0), and then evaluated with "the test inflow series" which has 400 sets of the mean (μ_1) and standard deviation (σ_1) values that are different from μ_0 and σ_0. The simulation results with test inflows were expressed in 2-D climate response functions with different measures of evaluation metrics. The overall flowchart of this case study is as in Figure 10.5.

10.3.2.1 Stochastic dynamic programming

SDP has been widely applied for reservoir operations (Stedinger et al., 1984; Kim & Palmer, 1997; Kim et al., 2007) because it can include various types of objective functions in its recursive equation and explicitly consider uncertainty in probabilistic terms. In this study, SDP was used to determine the monthly optimal release rules R_t^* with the discretized storage S_t^k and the monthly inflow Q_t as state variables. The probability

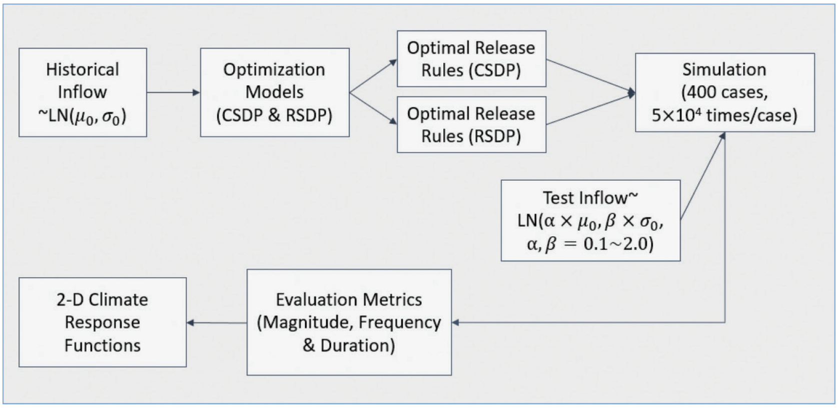

Figure 10.5 Overview of the first case study.

distribution of the monthly inflow was assumed to follow a lognormal distribution with historical statistics. The recursive equation and the constraints used in this study are shown in Equations (10.13–10.16).

$$f_t\left(S_t{}^k, Q_t\right) = \underset{R_t^*}{\text{Minimize}}\left[\left\{Z_t(\ ) + E_{Q_{t+1}|Q_t}\left[f_{t+1}\left(S_{t+1}{}^l, Q_{t+1}\right)\right]\right\}\right] \tag{10.13}$$

subject to

$$S_{t+1} = S_t + Q_t - R_t \tag{10.14}$$

$$R_t = \min\{\max[R_t, S_t - Q_t - S_{\max}], S_t - Q_t - S_{\min}\} \tag{10.15}$$

$$R_t \geq 0 \tag{10.16}$$

where $Z_t(\ )$ = the objective function to be minimized; $E_{Q_{t+1}|Q_t}[\ ]$ = the expectation over the specified conditional distribution of period $t + 1$ inflow, given the value of Q_t; $S_{\max}$ = the maximum reservoir storage (maximum water level); $S_{\min}$ = the minimum reservoir storage (dead storage level).

A common objective for the water supply is to minimize the difference between the demand, D_t, and the release, R_t, from the reservoir. This was selected as the main objective (σ_v in Equation (4)), with a quadratic form shown in Equation (10.17).

$$f_1 = \max\left(\frac{D_t - R_t}{D_t}, 0\right) \tag{10.17}$$

In Korea, the dam elevation at the end of drawdown season (i.e., June), S_T, is usually required to satisfy a specific standard elevation for the upcoming flood season, S_{target}. This elevation is expressed in Equation (10.18), and it was included in the above objective.

$$f_2 = \frac{S_T - S_{\text{target}}}{S_{\text{target}}} \tag{10.18}$$

However, note that the water supply objective is considerably more important than the flood preparation; hence, CSDP searches for the optimal solution with Equation (10.17) and then proceeds for the optimal solution with Equation (10.18). The objective function of the CSDP model was formulated as Equation (10.19). For further details of the SDP algorithm for reservoir operation, see Kim & Palmer (1997).

$$Z_{t,\text{CSDP}} = w_1 f_1^2 + w_2 f_2^2, \quad w_1 \gg w_2 \tag{10.19}$$

10.3.2.2 Robust stochastic dynamic programming

To formulate the shape of the feasibility penalty function $p(v_1, \ldots, v_S)$ in Equation (10.4), this study conducted multiple discussions with the stakeholders of the study basin, including dam operators and local government officials who constitute a structured public participation council. To help the stakeholders with different backgrounds in understanding the concept easily, this study modified the term in the constraint so that the state variables can take values below $S_{\min}$, and penalized the possible violation. The modified constraint in Equation (10.20) takes the value of $S_{\min,R}$ instead of $S_{\min}$, making the model use the water storage between $S_{\min}$ and $S_{\min,R}$ in case of emergency.

$$R_t = \min\{\max[R_t, S_t - Q_t - S_{\max}], S_t - Q_t - S_{\min,R}\} \tag{10.20}$$

Then, the violation of the original constraint (v_S in Equation (10.6)) is included in the feasibility penalty function with a new term f_3, where the magnitude of reservoir storage below the original constraint is incorporated into the penalty function.

$$f_3 = \max\left(\frac{S_{\min} - S_t}{S_{\min}}, 0\right) \tag{10.21}$$

Then, the objective function of the RSDP model was formulated as Equation (10.22) to derive the monthly optimal release of Boryeong Dam.

$$Z_{t,\mathrm{RSDP}} = w_1 f_1^2 + w_2 f_2^2 + w_1 f_3^2, \quad w_1 \gg w_2 \tag{10.22}$$

10.3.2.3 Simulation

As mentioned earlier, each tested inflow series has a different set of average and standard deviation values estimated from historical data (1998–2018), ($\mu_0 = 4.08\ \mathrm{m^3/s}$ and $\sigma_0 = 6.39\ \mathrm{m^3/s}$). The difference between the historical and tested series considers the non-stationary characteristic of climate change. In other words, the future may not be just a repetition of the past.

This study tested different values of α ranging from 0.1 to 2.0, where $\mu_1 = \alpha \times \mu_0$ (or $\sigma_1 = \alpha \times \sigma_0$). The range was selected based on the streamflow projections made by Seo and Kim (2018) which inputted 27 general circulation model (GCM) precipitation projections to a tank model (Sugawara, 1974) for the study basin. Figure 10.6 shows that the selected range ($\alpha = 0.1$–2.0) sufficiently covers the possible changes of average and standard deviation of the future streamflow for the study basin. This study tested 400 values of average and standard deviation (i.e., 20 μ_1 × 20 σ_1 values). For each of the 400 values, a streamflow series of 50,000 years was generated from an AR(1) time series model assuming that the streamflow data are lognormally distributed and their lag-1 autocorrelation structure remains the same in the future (Yoon et al., 2018).

The evaluation metrics selected in this study aim to account for the different measures of system performance and ensure that the measures considered important by the stakeholders are included in the metrics. It has been proven that, in theory, decision-makers with responsibility over dam operations are interested in a wide range of risks, including (i) risk of cost overruns (solution robustness), (ii) risk of system failure (reliability), (iii) magnitude of those failures (vulnerability), and (iv) risk of performance deterioration (sustainability) (Ray et al., 2014). Furthermore, Kim et al. (2019) studied this basin and observed that the duration of failure is more vulnerable to climate change and is also harder to predict than the other metrics designed to measure failure.

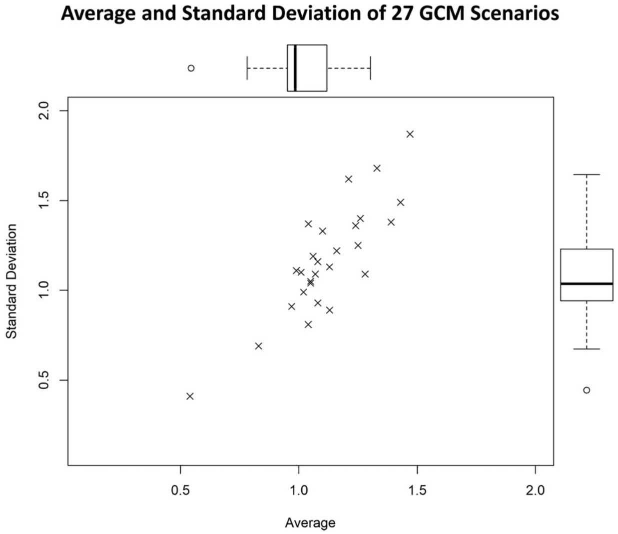

Figure 10.6 Average and standard deviation ratio based on 27 GCM outputs compared with the historical values.

By considering the opinions of multiple stakeholders and past research results, this study selected three metrics to express various aspects of potential risk effectively. The selected metrics represent the average magnitude, average frequency, and maximum duration based on the initial concepts suggested by Hashimoto et al. (1982). In Table 10.2, ADM_y was chosen to express the magnitude of the annual water deficit; ADF_y was chosen to express the average frequency of the annual water deficit; ReT_{max}

Table 10.2 Selected evaluation metrics

Metric	*Classification*	*Equation*	*Unit*
ADM_y	Magnitude of annual water deficit	$\frac{1}{T}\sum_{t=1}^{T}\max\left(\frac{D_t - R_t}{D_t}, 0\right)$	Ratio
ADF_y	Frequency of annual water deficit	$\frac{1}{T}\sum_{t=1}^{T} I_t\left(\frac{D_t - R_t}{D_t} > 0\right)$	Ratio
ReT_{max}	Duration of the longest failure	$\underset{t}{\mathrm{Max}}\{t_b - t_a\}$, for $\forall t_a$ s.t. $\frac{D_{t_a} - R_{t_a}}{D_{t_a}} > 0$ and $\frac{D_{t_a-1} - R_{t_a-1}}{D_{t_a-1}} \le 0$ and for first $t_b > t_a$ s.t. $\frac{D_{t_b} - R_{t_b}}{D_{t_b}} > 0$ and $\frac{D_{t_b-1} - R_{t_b-1}}{D_{t_b-1}} \le 0$	Months

ADM_y, average deficit magnitude; ADF_y, average deficit frequency; ReT_{max}, recovery time (maximum).

was selected as the longest time for the system to return from failure to success instead of average recovery time since the average recovery time showed no variation among different scenarios. All the metrics with higher values mean greater, more recurrent, and longer-lasting failure.

10.3.2.4 Results

Comparing the optimal release rules derived from CSDP and RSDP, it revealed notably significant features. Figure 10.7 compares the derived optimal October release rules averaged with respect to the inflow for both CSDP and RSDP. Due to the modified constraint, RSDP starts its release rule from a lower level (2.8 MCM) when compared with the starting point of CSDP (8.2 MCM). According to the optimal release rule for RSDP, it releases a slightly smaller amount than that released by CSDP when the storage is low and compensates for the low release when the storage is relatively higher.

It was also observed that, among the paths that were first neglected due to the harsh constraint in CSDP, several of them were made possible with RSDP where the constraint violation became possible. It was observed that, among all the possible state variables, 3.33% were below $S_{\min}$ (8.2 MCM) in the CSDP model, whereas 6.78% were below $S_{\min}$ in RSDP. The difference between these ratios indicates that RSDP transforms 3.45% of the possible state variables those were infeasible under CSDP into feasible state variables that could potentially be selected as an optimal solution.

In terms of the selected evaluation metrics, both CSDP and RSDP showed trade-offs under different conditions and different metrics. In terms of magnitude, the overall

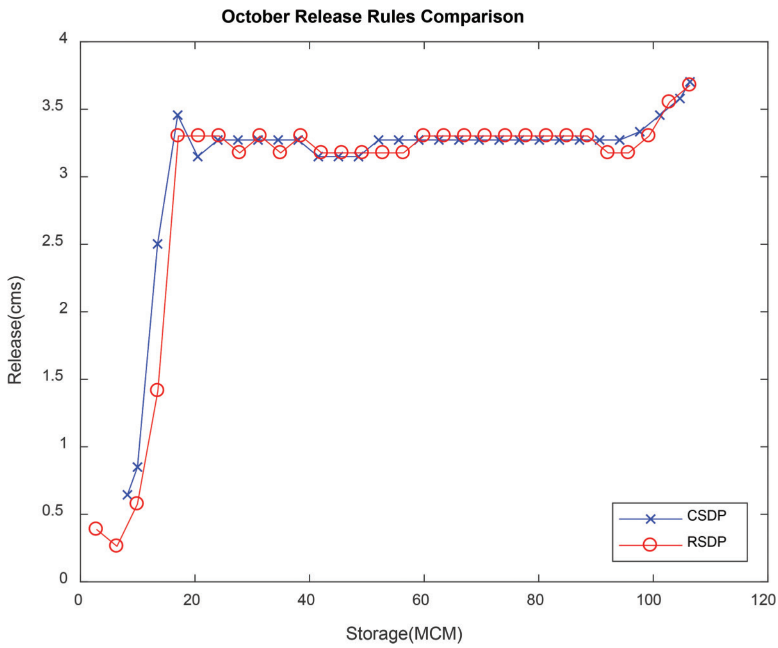

Figure 10.7 Release rule comparison of CSDP and RSDP (October).

results show no significant difference between models, but in extreme cases (decreased averages and increased standard deviations), RSDP performed slightly better than CSDP. However, in terms of frequency, RSDP performed better than CSDP overall, whereas CSDP performed better under cases with increased averages. In terms of the maximum duration of failure, CSDP performed better both in extreme cases and under increased averages of inflow. The contour plots of all the metrics are shown in Figure 10.8 and the results are summarized in Table 10.3.

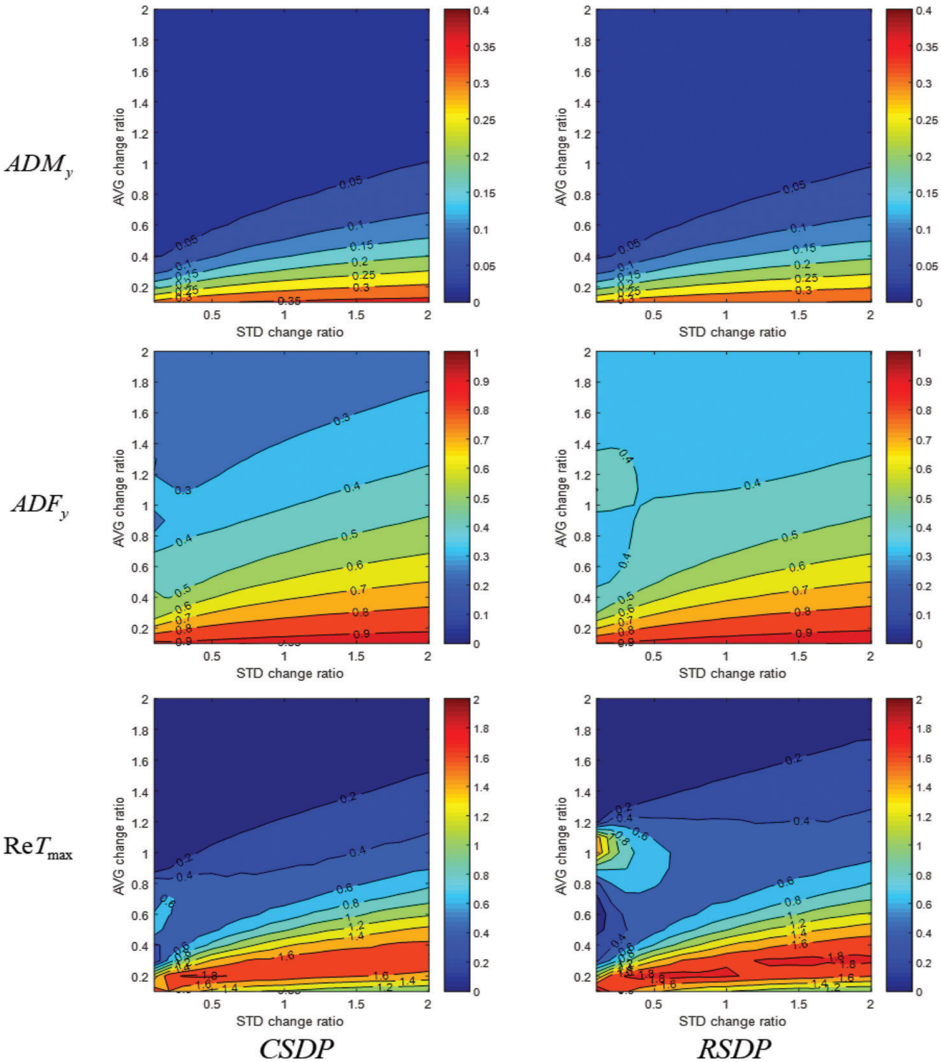

Figure 10.8 2-D contour plots of CSDP and RSDP models of each evaluation metric.

Table 10.3 Comparison of results

Metric	*Overall*	*Extreme cases*	*Increased averages*
ADM_y	Similar	RSDP > CSDP	Similar
AFM_y	RSDP > CSDP	Similar	CSDP > RSDP
ReT_{max}	CSDP > RSDP	CSDP > RSDP	Similar

Note: ">" indicates a better performance of the left model than that of the right model.

10.3.3 Water infrastructure planning with real option analysis

In this section, a case study of ROA that assesses water resources infrastructure planning for drought mitigation is introduced. This section is based on the study by Ihm et al. (2019), who developed a ROA model with the adaptive perspective to re-evaluate the economic feasibility of the Conduit Project described in Section 10.3.1.

10.3.3.1 Uncertainty in climate

Climate uncertainties are estimated as the occurrence probabilities of different drought conditions. Drought probabilities are calculated as follows: (i) Monthly streamflow series are synthetically generated by a periodic autoregressive moving average model. (ii) Monthly reservoir storage series is simulated by a reservoir model based on a drought contingency plan that was established for the target dam basin. (iii) The number of days in a year during which the reservoir stage is under a certain threshold level is calculated over the entire planning horizon. (iv) If the estimated number of days is greater than the value derived from a baseline drought scenario, the corresponding year is regarded as a severe drought year. If the estimated number is less than the baseline value, the corresponding year is regarded as a moderate drought year. If the estimated number is zero, the year is regarded as a normal year.

Figure 10.9 shows how the probabilities of drought occurrences are treated. As shown in Figure 10.9, three potential outcomes, i.e., normal (no drought), moderate, and severe drought, are postulated. The overall drought damage cost is calculated as the weighted sum of three different drought damage costs: i.e., overall drought damage cost = $P_{\_normal} \times C_{\_normal} + P_{\_moderate} \times C_{\_moderate} + P_{\_severe} \times C_{\_severe}$ (Ihm et al., 2019).

10.3.3.2 Drought damage cost

Drought damage cost can be calculated as the product of the unit price of water and the amount of water deficit. Assuming that the drought damage is proportional to the length of drought, Ihm et al. (2019) calculated the drought damage cost as presented in Equation (10.23).

$$C_d = ((W_d - W_s) \times WP_w) \times D_d \tag{10.23}$$

where C_d = the drought damage cost, W_d = the water demand per day, W_s = the water demand per day under the severe warning stage, WP_w = the unit price of water, and D_d = the average number of days under the severe warning stage (Ihm et al., 2019).

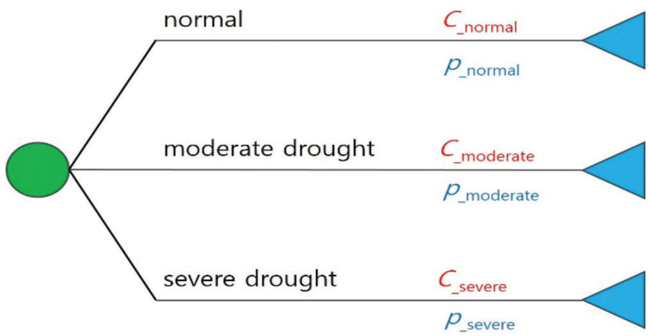

Figure 10.9 Evaluation of the uncertainty factor by postulating multiple potential conditions. *c*: drought damage cost for each condition; *p*: the probability of drought for each condition (Ihm et al., 2019).

Furthermore, the stress cost, which is an additional damage defined as an amount of mental stress due to restricted water supply, can be considered as a monetary value. Then, the term WP_w in Equation (10.23) becomes $(WP_w + S_w)$ where S_w is the unit price of mental stress due to water supply restriction.

10.3.3.3 ROA modeling using the decision tree approach

Three options, "invest," "delay," and "abort," are used in the case study. With the investment option, a project is installed immediately and operated until the end of the planning horizon. Conversely, if the delay option is chosen, the installation is postponed and re-evaluated in the following time step. Finally, when the installation and delay of the project are no longer economically feasible, the abort option is suggested. With the abort option, the project is canceled so that no further investment is considered during the planning horizon. Figure 10.10 demonstrates the decision tree generated for ROA based on the three options.

With the calculated ENPV value for each year, the abort option is suggested when the OP value is zero. Otherwise, when the ENPV is greater than the NPV (i.e., the OP value is positive), the invest (delay) option is chosen if the option value (sum of the net benefit of the option) of invest is greater (lesser) than the option value of delay. When the delay option is selected, all the options are available for the next year. The optimal option is determined based on the previously mentioned selection mechanism. The outcomes of the existing feasibility study on the Conduit Project (KDI, 2016) are compared with those of the ROA model to evaluate the developed ROA model with the decision tree approach.

10.3.3.4 Results

Ihm et al. (2019) re-evaluated the economic feasibility of the Conduit Project and discussed that the abort option could be the optimal choice for the lowest economic loss on the project. In reality, the Conduit Project caused a nationwide controversy due

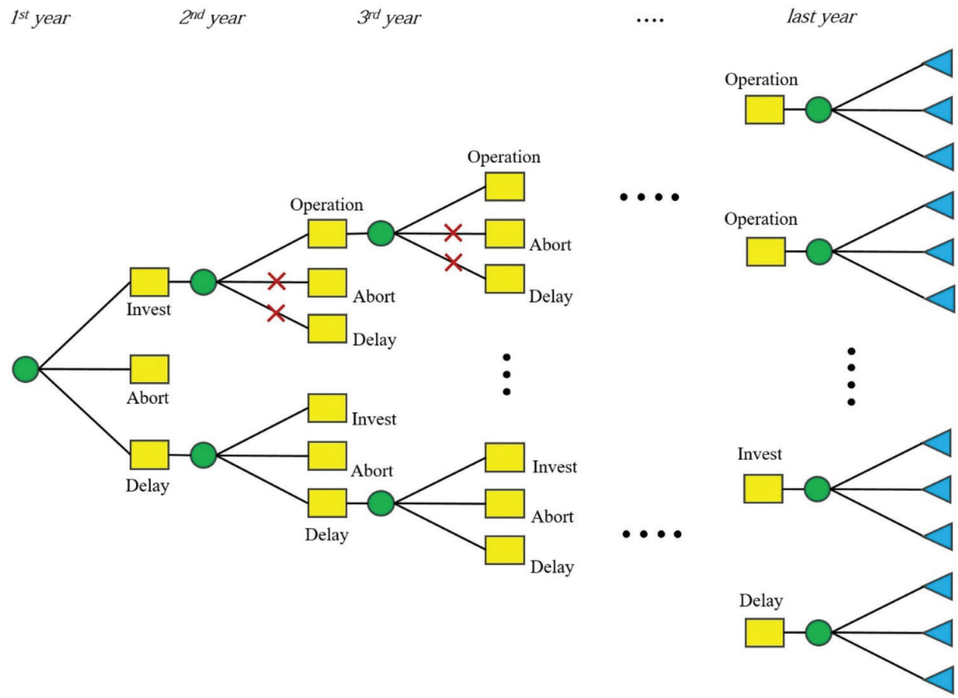

Figure 10.10 Decision tree based on the three options: invest, delay, and abort (Ihm et al., 2019).

to its enormous capital cost despite its advantage in alleviating water shortage immediately. Table 10.4 presents the economic feasibility results evaluated using various BCA models. As the total expenses of the Conduit Project were greater than the initial estimation of the project, all the BCA results (total benefit in Table 10.4) were negative, which indicates that the project was not economically feasible at the time of construction.

Table 10.4 Results of BCA for Boryeong Dam Conduit Project (Ihm et al., 2019)

Models	DCF_{KDI} *(baseline)*[a]	DCF_{UNC}[b]	ROA_{UNC}[c]	DCF_{UNC_S}[d]	ROA_{UNC_S}[e]
Total benefit (million USD)	NPV −83.4	NPV −89.3	ENPV −2.0	NPV −94.3	ENPV −79.5

[a] The traditional DCF model conducted by KDI (2016).
[b] The DCF model reflecting climate uncertainty.
[c] The ROA model reflecting both climate uncertainty and flexibility in the decision-making process.
[d] The same model as (b), but the stress cost is added to drought damage.
[e] The same model as (c), but the stress cost is added to drought damage.

Compared with the DCF_{KDI}, the NPV of the DCF_{UNC} model decreased by approximately 7% (5.9 million USD). This decrease indicates that the economic feasibility of the project was over-estimated with no consideration of uncertainty in climate. Conversely, the ROA_{UNC} (which selects the abort option), was the most valuable economically (i.e., the optimal option), although the ENPV was still negative. This suggests that the invest option is not better than the abort option. The OP obtained using the abort option was 81.4 USD, (= (−2.0) − (−83.4)). Including the stress cost in the drought damage, in the case of DCF (by comparing the difference between the total benefits of DCF_{UNC} and DCF_{UNC_S}), the impact of the additional stress cost on the economic feasibility was relatively less, as the conduit can reduce the water shortage. Conversely, there was a notable difference between the total benefits (ENPV) of ROA_{UNC} and ROA_{UNC_S}. This difference indicates that the additional stress cost due to water shortage was added to the ENPV by selecting the abort option for the optimal decision (Ihm et al., 2019).

Furthermore, sensitivity analyses were implemented to find other conditions that can increase the economic worth of the current project (Ihm et al., 2019). In this study, the one-factor-at-a-time (OFAT) method (Czitrom, 1999), which is one of the most commonly used methods, is applied for the sensitivity analysis. Theoretically, OFAT experiments change only one variable at a time while maintaining others fixed. Figure 10.11 shows the results of the sensitivity analyses of the case study.

The ROA model determined that the invest option would be better than the abort option when the probability of severe drought increases by approximately 20%. This indicates that the Conduit Project can be economically feasible if the probability of severe drought is increased by 20%. Moreover, if the total project expense decreases by approximately 27%, the invest option becomes economically feasible. Thus, the sensitivity analyses provide insights into a possible strategy for maximizing the economic feasibility of the project (Ihm et al. 2019).

10.4 Conclusions

Adapting to climate change is often a challenge in decision-making processes, and the water sector is not an exception. Several studies during the last two decades demonstrated that the decision-making strategy under climate change should be robust and adaptive to consider effectively two key characteristics of climate change: non-stationarity and large (or sometimes deep) uncertainty. The robust strategy is satisfactory over a wide range of uncertainty, whereas the adaptive strategy updates itself continuously with its performance and new information. This chapter introduced two well-known decision-making methods: robust optimization for the robust perspective, and ROA for the adaptive perspective.

Robust optimization involves saving infeasible scenarios that violate the constraints being considered by incorporating a penalty function, which is a function of the violation degree for each scenario. This study developed the RSDP model for robust optimization and applied it to the monthly reservoir operations of a single dam in Korea. The derived release policies were then tested with the synthetic inflow series, which has different averages and standard deviations from the ones estimated when the policy was derived; these changes can reflect the non-stationarity of climate change. It was observed that the RSDP model performed better than the CSDP model in terms of

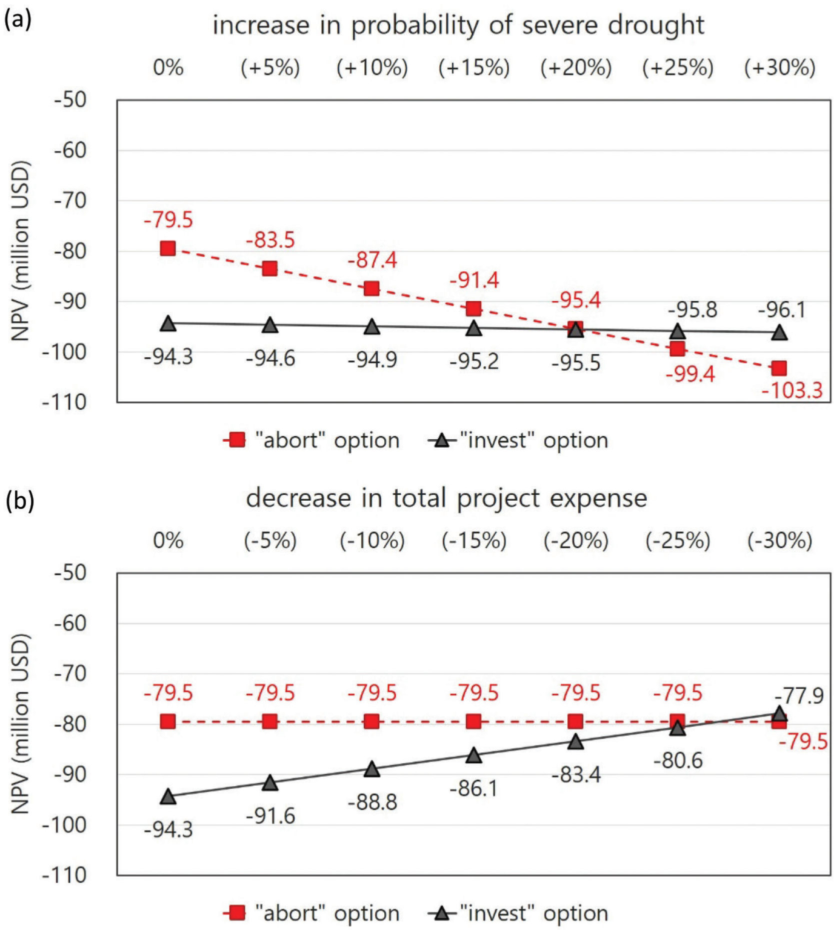

Figure 10.11 Results of sensitivity analyses: (a) probability of severe drought and (b) total project expense (Ihm et al., 2019).

frequency, and magnitude under extreme cases with a dramatically decreased average of inflow.

ROA expands the traditional BCA by incorporating options and scenarios into its evolutionary procedure. Each alternative can use multiple options; the decision-maker updates her/his decision adapting to the decision result made in the previous stage. This study applied a ROA model to a water infrastructure planning project for drought mitigation in Korea and compared the result of ROA with that of the existing government analysis, which considered neither uncertainty nor flexibility (i.e., OP). The case study demonstrated that the project being considered is not economically feasible when it is constructed but it could become feasible if the probability of severe drought increases by 20%.

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References

Ben-Haim, Y. (2001). Info-gap value of information in model updating. *Mechanical Systems and Signal Processing*, 15 (3), 457–474.

Bloom, E. (2014). Changing midstream: Providing decision support for adaptive strategies using Robust Decision Making: Applications in the Colorado River Basin (*Doctoral dissertation*, The Pardee RAND Graduate School).

Borison, A., Hamm, G., Farrier, S., & Swier, G. (2008). *Real options and urban water resource planning in Australia*. Sydney, Water Services Association of Australia.

Buurman, J. & Babovic, V. (2016). Adaptation pathways and real options analysis: An approach to deep uncertainty in climate change adaptation policies. *Policy and Society*, 35 (2), 137–150. Available from: doi: 10.1016/j.polsoc.2016.05.002 [Accessed October 16, 2019].

Buurman, J., Zhang, S., & Babovic, V. (2009). Reducing risk through real options in systems design: The case of architecting a maritime domain protection system. *Risk Analysis*, 29 (3), 366–379. Available from: doi: 10.1111/j.1539-6924.2008.01160.x [Accessed October 16, 2019].

Chow, J. Y., & Regan, A. C. (2011). Network-based real option models. *Transportation Research Part B: Methodological*, 45 (4), 682–695.

Chung, E. S., & Kim, Y. (2014). Development of fuzzy multi-criteria approach to prioritize locations of treated wastewater use considering climate change scenarios. *Journal of Environmental Management*, 146, 505–516.

Czitrom, V. (1999). One-factor-at-a-time versus designed experiments. *The American Statistician*, 53 (2), 126–131. Available from: doi: 10.1080/00031305. 1999.10474445 [Accessed October 16, 2019].

De Neufville, R. (2003). Real options: dealing with uncertainty in systems planning and design. *Integrated Assessment*, 4 (1), 26–34

De Neufville, R. & Scholtes, S. (2011). *Flexibility in engineering design*. Cambridge, MIT Press.

Deng, Y., Cardin, M. A., Babovic, V., Santhanakrishnan, D., Schmitter, P., & Meshgi, A. (2013). Valuing flexibilities in the design of urban water management systems. *Water Research*, 47 (20), 7162–7174. Available from: doi:10.1016/j.watres.2013.09.064 [Accessed October 16, 2019].

Dessai, S. & Hulme, M. (2007). Assessing the robustness of adaptation decisions to climate change uncertainties: A case study on water resources management in the East of England. *Global Environmental Change*, 17 (1), 59–72.

Dessai, S. (2005). Robust adaptation decisions amid climate change uncertainties (*Doctoral dissertation*, University of East Anglia).

Dessai, S., Hulme, M., Lempert, R., & Pielke Jr, R. (2009). Climate prediction: A limit to adaptation. In: Adger, W. N., Lorenzoni, I., & O'Brien, K. L. (eds.) *Adapting to climate change: thresholds, values, governance*. Book information, pp. 64–78.

Erfani, T., Pachos, K., & Harou, J. (2018). Real-options water supply planning: Multistage scenario trees for adaptive and flexible capacity expansion under probabilistic climate change uncertainty. *Water Resources Research*, 54 (7), 5069–5087. Available from: doi: 10.1029/2017WR021803 [Accessed October 16, 2019].

Gersonius, B., Ashley, R., Pathirana, A., & Zevenbergen, C. (2012). Climate change uncertainty: Building flexibility into water and flood risk. *Climatic Change*, 116, 411–423. Available from: doi: 10.1007/ s10584-012-0494-5 [Accessed October 16, 2019].

Hashimoto, T., Loucks, D. P., & Stedinger, J. R. (1982). Robustness of water resources systems. *Water Resources Research*, 18 (1), 21–26.

Herman, J. D., Reed, P. M., Zeff, H. B., & Characklis, G. W. (2015). How should robustness be defined for water systems planning under change? *Journal of Water Resources Planning and Management*, 141 (10), 04015012.

Ihm, S. H., Seo, S. B., & Kim, Y.-O. (2019). Valuation of water resources infrastructure planning from climate change adaptation perspective using real option analysis. *KSCE Journal of Civil Engineering*, (2012), 1–9. Available from: doi: 10.1007/s12205-019-1722-6 [Accessed October 16, 2019].

Jeuland, M. & Whittington, D. (2014). Water resources planning under climate change: Assessing the robustness of real options for the Blue Nile. *Water Resources Research*, 50, 2086–2107. Available from: doi: 10.1002/2013WR013705 [Accessed October 16, 2019].

KDI (Korea Development Institute) (2016). *Cost sharing plan of Boryeong Dam conduit project*. Sejong, Korea, Public Invest Management Center of KDI.

Kim, G. J., Seo, S. B., & Kim, Y.-O. (2019). Elicitation of drought alternatives based on Water Policy Council and the role of Shared Vision Model. *Journal of Korea Water Resources Association*, 52 (6), 429–440.

Kim, Y.-O. & Chung, E. S. (2014). An index-based robust decision making framework for watershed management in a changing climate. *Science of the Total Environment*, 473, 88–102.

Kim, Y.-O. & Chung, E. S. (2015). Robust prioritization of climate change adaptation strategies using the VIKOR method with objective weights. *Journal of the American Water Resources Association*, 51 (5), 1167–1182.

Kim, Y.-O. & Chung, E. S. (2017). Adaptation to climate change: Decision-making. In: Kolokytha, E., Oishi, S., & Teegavarapu, R. S. V. (eds.) *Sustainable water resources planning and management under climate change*. Singapore, Springer. pp. 189–221.

Kim, Y.-O., Eum, H.-I., Lee, E.-G., & Ko, I.H. (2007). Optimizing operational policies of a Korean multireservoir system using sampling stochastic dynamic programming with ensemble streamflow prediction. *Journal of Water Resources Planning and Management*, 133 (1), 4–14.

Kim, Y.-O. & Palmer, R. N. (1997). Value of seasonal flow forecasts in Bayesian stochastic programming. *Journal of Water Resources Planning and Management*, 123 (6), 327–335.

Kjaerland, F. (2007). A real option analysis of investments in hydropower: The case of Norway. *Energy Policy*, 35 (11), 5901–5908. Available from: doi: 10.1016/j.enpol.2007.07.021 [Accessed October 16, 2019].

Lander, D. M. & Pinches, G. E. (1998). Challenges to the practical implementation of modeling and valuing real options. *The Quarterly Review of Economics and Finance*, 38 (3), 537–567.

Lempert, R. J. (2003). *Shaping the next one hundred years: new methods for quantitative, long-term policy analysis*. Santa Monica, CA, Rand Corporation.

Lempert, R., Nakicenovic, N., Sarewitz, D., & Schlesinger, M. (2004). Characterizing climate-change uncertainties for decision-makers. An editorial essay. *Climatic Change*, 65 (1), 1–9.

Lempert, R., Kalra, N., Peyraud, S., Mao, Z., Tan, S. B., Cira, D., & Lotsch, A. (2013). *Ensuring robust flood risk management in Ho Chi Minh City*. Washington, DC, The World Bank.

Lempert, R. & Groves, D. (2010). Identifying and evaluating robust adaptive policy responses to climate change for water management agencies in the American west. *Technological Forecasting and Social Change*, 77 (6), 960–974.

Matrosov, E., Woods, A., & Harou, J. (2013). Robust decision making and info-gap decision theory for water resource system planning. *Journal of Hydrology*, 494, 43–58.

Mulvey, J. M., Vanderbei, R. J., & Zenios, S. A. (1995). Robust optimization of large-scale systems. *Operations Research*, 43 (2), 264–281.

Myers, S. C. (1977). Determinants of corporate borrowing. *Journal of Financial Economics*, 5 (2), 147–175.

Pan, L., Housh, M., Liu, P., Cai, X., & Chen, X. (2015). Robust stochastic optimization for reservoir operation. *Water Resources Research*, 51 (1), 409–429.

Park, T., Kim, C., & Kim, H. (2014). Valuation of drainage infrastructure improvement under climate change using real options. *Water Resources Manager*, 28 (2), 445–457. Available from: doi: 10.1007/s11269-013-0492-z [Accessed October 16, 2019].

Ranger, N., Millner, A., Dietz, S., Fankhauser, S., Lopez, A., & Ruta, G. (2010). Adaptation in the UK: a decision-making process. *Environment Agency*, 9, 1–62.

Ray, P. A., Watkins Jr, D. W., Vogel, R. M., & Kirshen, P. H. (2014). Performance-based evaluation of an improved robust optimization formulation. *Journal of Water Resources Planning and Management*, 140 (6), 04014006.

Ryu, Y., Kim, Y.-O., Seo, S. B. & Seo, I. W. (2017). Application of real option analysis for planning under climate change uncertainty: A case study for evaluation of flood mitigation plans in Korea. *Mitigation and Adaptation Strategies for Global Change*, 23 (6), 803–819. Available from: doi: 10.1007/s11027-017-9760-1 [Accessed October 16, 2019].

Stedinger, J., Sule, B., & Loucks, D. (1984). Stochastic dynamic programming models for reservoir operation optimization. *Water Resources Research*, 20 (11), 1499–1505.

Seo, S. B. & Kim, Y.-O. (2018). Impact of spatial aggregation level of climate indicators on a national-level selection for representative climate change scenarios. *Sustainability*, 10 (7), 2409. Available from: doi: 10.3390/su10072409 [Assessed October 16, 2019].

Steinschneider, S. & Brown, C. (2012). Dynamic reservoir management with real-option risk hedging as a robust adaptation to nonstationary climate. *Water Resources Research*, 48 (5). Available from: doi: 10.1029/ 2011WR011540 [Accessed October 16, 2019].

Sugawara, M. (1974). Tank model and its application to Bird Creek, Wollombi Brook, Bikin River, Kitsu River, Sanaga River and Nam Mune. *Research Notes of the National Research Center for Disaster Prevention*, 11, 1–64.

Trigeorgis, L. (1993). Real options and interactions with financial flexibility. *Financial Management*, 22 (3), 202–224. Available from: doi: 10.2307/3665939 [Accessed October 16, 2019].

Trigeorgis, L. (1996). *Real options: Managerial flexibility and strategy in resource allocation*. Cambridge, MIT Press.

Walker W. E., Haasnoot, M., & Kwakkel, J. H. (2013) Adapt or perish: A review of planning approaches for adaptation under deep uncertainty. *Sustainability*, 5 (3), 955–979. Available from: doi: 10.3390/su5030955 [Accessed October 16, 2019].

Watkins Jr, D. W. & McKinney, D. C. (1997). Finding robust solutions to water resources problems. *Journal of Water Resources Planning and Management*, 123 (1), 49–58.

Yoon, H. N., Seo, S. B., & Kim, Y.-O. (2018). Development of Robust-SDP for improving dam operation to cope with non-stationarity of climate change. *Journal of Korea Water Resources Association*, 51 (11), 1135–1148.

Zhang, S. X. & Babovic, V. (2011). An evolutionary real options framework for the design and management of projects and systems with complex real options and exercising conditions. *Decision Support Systems*, 51 (1), 119–129. Available from: doi: 10.1016/j.dss. 2010.12.001 [Accessed October 16, 2019].